



***Progressive Education Society's***  
**Modern College of Arts, Science and Commerce, (Autonomous)**  
**Shivajinagar, Pune-411005.**

# **OJT PROJET REPORT**

**JUNE 2026**

**A Project Report Submitted for The On-Job Training (OJT)**  
**Of Master of Arts/Science/Commerce/ Vocational In <<specialization>>**

***Name-Bedmutha Jiya Pravin***

***Class- M.Sc. Statistics (I)***

***Roll No.-2533304***

***Date of Submission-June 16, 2026***

## INDEX

| Sr. No. | Particulars                                   | Page No. |
|---------|-----------------------------------------------|----------|
| 1       | Certificate                                   | 1        |
| 2       | Attendance Sheet                              | 2        |
| 3       | Application for On Job Training               | 3        |
| 4       | Acknowledgement                               | 4        |
| 5       | Abstract                                      | 5        |
| 6       | Problem Definition                            | 6        |
| 7       | Need for the System                           | 7        |
| 8       | Scope of the Proposed System                  | 8        |
| 9       | Objectives of the Proposed System             | 9        |
| 10      | Requirement Analysis / Analysis Specification | 10       |
| 11      | Literature Review                             | 11       |
| 12      | Design Specifications                         | 12       |
| 13      | Methodology                                   | 13       |
| 14      | Phase 1 Analysis and Findings                 | 14       |
| 15      | Phase 2 Data Collection and Analysis          | 15       |
| 16      | Product Optimization and Ranking Analysis     | 16       |

|    |                                    |    |  |
|----|------------------------------------|----|--|
| 17 | Optimum Product Identification     | 17 |  |
| 18 | Implementation Strategies          | 18 |  |
| 19 | Limitations and Observations       | 19 |  |
| 20 | Conclusion and Future Enhancements | 20 |  |
| 21 | User Manual                        | 21 |  |
| 22 | References                         | 21 |  |

### **On Job Training Completion Certificate**

**This is to confirm that Mr./Ms. Jiya Pravin Bedmutha of Class M.Sc. I (Statistics) Division A having roll number 2533304 has successfully completed the On Job Training in the Subject/Course Name '*On Job Training*' prescribed for Semester II of the academic year 2025-2026.**

**The duration of the On Job Training in our organization was from 1 May 2026 to 15 June 2026.  
She has worked 4 hours per day.**

**Her performance during the On Job Training in our Organization was Excellent / Good/  
Satisfactory/ unsatisfactory.**

**We are relieving her on date 15 June after her office hours.**

**We wish her all the best for her future endeavors.**

**Modern College of Arts, Science and Commerce (Autonomous) Shivajinagar, Pune 5**

**Attendance Sheet**

**Name of the Student:** Bedmutha Jiya Pravin

**Class & Division:** M.Sc. (I) Statistics

**Name of the company/ Institute:** PURSULLENCE GLOBAL BUSINESS SOLUTIONS LLP, INDIA.

| Sr.No | Date       | Time in | Time out | Sign of Student | Sign of external supervisor | Remark |
|-------|------------|---------|----------|-----------------|-----------------------------|--------|
| 1     | 06-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 2     | 07-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 3     | 08-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 4     | 09-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 5     | 10-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 6     | 11-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 7     | 13-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 8     | 14-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 9     | 15-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 10    | 16-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 11    | 17-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 12    | 18-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 13    | 20-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 14    | 21-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 15    | 22-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 16    | 24-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 17    | 25-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 18    | 27-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |
| 19    | 28-05-2024 | 10:00AM | 2:00PM   |                 |                             |        |

|    |            |         |        |  |  |  |
|----|------------|---------|--------|--|--|--|
| 20 | 29-05-2024 | 10:00AM | 2:00PM |  |  |  |
| 21 | 30-05-2024 | 10:00AM | 2:00PM |  |  |  |
| 22 | 31-05-2024 | 10:00AM | 2:00PM |  |  |  |
| 23 | 01-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 24 | 03-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 25 | 04-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 26 | 05-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 27 | 06-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 28 | 07-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 29 | 08-06-2024 | 10:00AM | 2:00PM |  |  |  |
| 30 | 10-06-2024 | 10:00AM | 2:00PM |  |  |  |

***Progressive Education Society's***

**Modern College of Arts, Science and Commerce (*Autonomous*)**

**Shivajinagar, Pune 5**

**To,**

\_\_\_\_\_  
\_\_\_\_\_

**Subject: Application for the 'On Job Training' in your organization**

Dear Sir / Madam,

Progressive Education Society's, Modern College of Arts, Science and Commerce (*Autonomous*), Shivajinagar, Pune 5 was established in 1970. It is permanently affiliated to Savitribai Phule Pune University, Pune. The College has been awarded Autonomous Status by University Grants Commission, New Delhi since 2019 and Empowered Autonomous Status by SPPU, Pune. The College is accredited by NAAC Bangalore with A+ Grade.

As per the guidelines of Govt. of Maharashtra, the Autonomous Colleges need to implement National Education Policy 2020 (NEP 2020) from the academic year 2023-24. We are happy to share with you that we have successfully implemented NEP 2020 from the academic year 2025-26. Under NEP 2020, for the Post Graduate courses, the PG Part-1 students have to undergo 'On Job Training' course of 4 credits as a part of their summer vacation. The student has to complete 120 hours of working in the duration of 1.5 months i.e. from 1<sup>st</sup> May to 15<sup>th</sup> June 2026. The student may come for the 'On Job Training' daily 4/6/8 office hours to complete the designated 120 hours of the training.

Our student \_\_\_\_\_ roll number \_\_\_\_\_ wishes to apply for the 'On Job Training' in your esteemed organization. The internal guide for him/her is \_\_\_\_\_ (Mobile No: \_\_\_\_\_).

You are appealed to maintain the daily attendance record of the student, signed by the authority from your organization to whom he / she will be reporting. At the end of the training, the attendance report may be handed over to the student along with the completion certificate. (Both the formats are attached herewith).

**Internal Guide**  
**(PG Section)**

**Head of the Department**

**Vice Principal**

## **ACKNOWLEDGEMENT**

I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of my On-Job Training (OJT) Project titled “Personal Loan: Financial Services.”

I am thankful to the Principal, Vice Principal, Head of the Department, and all faculty members of the Department of Statistics, Modern College of Arts, Science and Commerce (Autonomous), Shivajinagar, Pune, for providing me with an opportunity to undertake this project as a part of the curriculum under the National Education Policy (NEP 2020).

I extend my heartfelt thanks to my project guide and mentors for their valuable guidance, encouragement, and constructive suggestions throughout the project. Their support helped me understand the practical application of statistical and analytical techniques in the financial services sector.

I am also grateful to all the respondents who participated in the survey and provided valuable information for the successful completion of this study. Their cooperation made it possible to collect meaningful data and derive useful insights.

Finally, I express my sincere gratitude to my parents, friends, and well-wishers for their constant support, motivation, and encouragement during the entire project work.



## **ABSTRACT**

The present study, “Personal Loan: Financial Services,” focuses on understanding customer preferences regarding personal loan products and identifying the factors that significantly influence their borrowing decisions. In today's competitive financial environment, banks and financial institutions must design customer-centric products that align with customer expectations and market requirements.

The study was conducted using a survey-based methodology. Data were collected through structured questionnaires and processed using Microsoft Excel and the Pursuence AI-Predictive Analytics Tool.

Customer responses were converted into binary format and analyzed to determine the relative importance of various personal loan attributes such as disbursement time, documentation requirements, foreclosure facility, repayment tenure, insurance benefits, and interest structure.

The analysis revealed significant variation in customer preferences and identified Disbursement Time as the most influential factor affecting personal loan selection. Using predictive analytics techniques, the study evaluated multiple combinations of loan features and determined the optimum product configuration preferred by customers. The results demonstrate that customers value convenience, flexibility, transparency, and affordability while selecting personal loan products.

The findings of this project provide useful insights for financial institutions in designing customer-focused loan products and highlight the importance of data-driven decision-making in modern financial services. The study also provided practical exposure to survey design, data collection, statistical analysis, predictive analytics, and business decision-making processes.

**Keywords:** Personal Loan, Customer Preference Analysis, Predictive Analytics, Financial Services, Survey Research, Product Optimization.

## **Problem Definition**

The personal loan market has become highly competitive, with multiple financial institutions offering similar loan products. Customers evaluate several factors such as processing fees, disbursement time, documentation requirements, repayment flexibility, insurance benefits, and interest structures before selecting a personal loan.

However, financial institutions often face challenges in identifying which product features are most important to customers and which combination of features can maximize customer acceptance. This project aims to identify the key factors influencing customers' personal loan preferences and to optimize the loan product design using data-driven analysis.

Through customer surveys and predictive analytics, the study seeks to determine the most preferred loan attributes and recommend an optimized personal loan offering that aligns with customer expectations.

## **Need for the System**

In the modern financial services sector, understanding customer preferences is essential for designing competitive and customer-centric loan products. Traditional decision-making based on assumptions may not accurately reflect customer requirements and can lead to lower customer satisfaction and reduced product acceptance.

There is a need for a systematic and analytical approach to capture customer preferences and transform them into actionable business insights. The AI-Predictive Analytics Tool used in this project helps identify the most influential factors affecting personal loan selection and evaluates customer responses scientifically.

The system enables financial institutions to:

1. Identify the most important factors influencing personal loan selection.
2. Understand customer preferences regarding loan features and benefits.
3. Optimize product offerings based on data-driven insights.
4. Improve customer satisfaction and product acceptance.
5. Support strategic decision-making in personal loan product design.
6. Enhance competitiveness in the financial services market.

Therefore, the implementation of an analytics-based decision support system is necessary to develop customer-focused personal loan products and improve overall business performance.

## **Scope of the Proposed System**

The scope of this project is limited to understanding customer preferences related to personal loan products and optimizing loan features using predictive analytics techniques. The study focuses on collecting customer responses through structured questionnaires and analyzing the data using the AI-Predictive Analytics Tool.

The project covers the evaluation of major personal loan attributes such as processing fees, loan disbursal time, documentation requirements, loan protection insurance, repayment tenure, and interest structure. The analysis helps identify the most preferred features and the optimal combination of loan offerings from a customer perspective.

The scope of the system includes:

1. Collection of customer preference data through surveys.
2. Conversion of collected responses into binary (0–1) format for analysis.
3. Identification of key factors influencing personal loan selection.
4. Statistical and predictive analysis of customer preferences.
5. Optimization of personal loan product features.
6. Generation of business recommendations based on analytical findings.

The project is intended for academic and analytical purposes and does not involve actual loan approval, credit assessment, customer verification, or financial transactions.

## **Objectives of the Proposed System**

The primary objective of this project is to analyze customer preferences regarding personal loan products and develop an optimized loan offering strategy using data-driven decision-making techniques.

The specific objectives of the project are:

1. To identify the factors that influence customers while selecting a personal loan.
2. To collect and analyze customer preference data through structured questionnaires.
3. To convert survey responses into a binary data format suitable for predictive analytics.
4. To determine the relative importance of different personal loan features.
5. To identify customer-preferred alternatives for each loan feature.
6. To optimize the personal loan product design using analytical tools and statistical techniques.
7. To provide actionable recommendations for improving customer acceptance and satisfaction.
8. To understand the practical application of predictive analytics in financial services.
9. To support data-driven business strategy formulation for personal loan products.
10. To gain hands-on experience in survey design, data collection, data processing, and business analytics.

## **Requirement Analysis / Analysis Specification**

The project required the collection and analysis of customer preference data related to personal loan products. The following requirements were identified for the successful completion of the project:

### **1. Data Requirements**

- Collection of primary data through structured questionnaires.
- Customer responses related to personal loan preferences.
- Data suitable for binary (0–1) conversion and analytical processing.

### **2. Analytical Requirements**

- Identification of important factors influencing personal loan selection.
- Evaluation of customer preferences for various loan features.
- Analysis and optimization using the AI-Predictive Analytics Tool.
- Generation of data-driven business insights and recommendations.

### **3. System Requirements**

- Google Forms for questionnaire creation and response collection.
- Microsoft Excel for data cleaning, preprocessing, and binary conversion.
- AI-Predictive Analytics Tool for statistical analysis and feature optimization.
- Computer system with spreadsheet and analytical software support.

### **4. Output Requirements**

- Identification of key factors affecting personal loan selection.
- Ranking of customer-preferred loan features.
- Optimized personal loan product strategy.
- Business recommendations based on customer preferences.

The above requirements formed the basis for data collection, analysis, optimization, and interpretation of results in the present study.

## **Literature Review**

The financial services industry has witnessed significant growth in the use of data analytics and customer-centric product design. Financial institutions increasingly rely on customer preference studies and predictive analytics techniques to understand customer behavior and improve their products and services.

Several studies have indicated that factors such as interest rates, processing fees, loan disbursement speed, documentation requirements, repayment flexibility, and insurance benefits play an important role in influencing customers' borrowing decisions. Customers generally prefer loan products that offer convenience, transparency, affordability, and flexibility.

With the advancement of technology, banks and financial institutions have started using data-driven decision-making approaches to optimize their loan offerings. Predictive analytics enables organizations to analyze customer responses, identify preference patterns, and design products that better align with customer expectations.

Customer preference analysis is widely used in marketing and financial services to determine the relative importance of product attributes and to identify the most preferred combinations of features. Such analytical techniques help organizations improve customer satisfaction, increase product acceptance, and gain a competitive advantage in the market.

The present study applies these concepts in the context of personal loan products. By collecting customer responses and analyzing them using the AI-Predictive Analytics Tool, the project aims to identify key factors affecting personal loan selection and recommend optimized loan product strategies based on customer preferences.

## **Design Specifications**

The project was designed using a survey-based analytical approach to understand customer preferences regarding personal loan products. The study was conducted in two phases.

### **Phase 1 Design**

A structured questionnaire was developed to identify the most important factors influencing personal loan selection. Respondents were asked to select up to five factors from a list of eight personal loan attributes, including processing fees, disbursement time, foreclosure facilities, partial-payment facilities, documentation requirements, repayment tenure, interest structure, and loan protection insurance.

### **Phase 2 Design**

Based on the results of Phase 1, the most significant factors were selected for further analysis. A second questionnaire was designed containing multiple alternative offerings for each selected factor. Respondents were required to choose their preferred option for each feature.

### **Data Processing Design**

The collected responses were exported to Microsoft Excel and converted into binary (0–1) format. The processed data was then analyzed using the AI-Predictive Analytics Tool to identify customer preferences and optimize product features.

### **Output Design**

The system generated analytical outputs including feature prioritization, preference rankings, statistical measures, and optimized business recommendations. These outputs were used to identify the most preferred personal loan product configuration from a customer perspective.



## Method

The project was carried out using a survey-based analytical methodology consisting of two phases.

### Step 1: Project Selection

The project area assigned was Financial Services – Personal Loan. The objective was to identify customer preferences and optimize personal loan product features.

### Step 2: Phase 1 Data Collection

A structured questionnaire was administered to respondents. They were asked to select the factors they considered most important while choosing a personal loan. Responses were collected from approximately 100 respondents.

### Step 3: Data Preparation

The collected responses were converted into binary (0–1) format using Microsoft Excel. The prepared dataset was checked for consistency and accuracy.

## Analysis of Key Factors Influencing Personal Loan Selection

This survey is conducted as part of an academic project to understand the key factors considered while choosing a personal loan.

All responses will be used strictly for educational purposes, and your information will remain confidential.

...

Which of the following factors do you consider most important while selecting a personal loan?  
(Please select up to 5 options only)

व्यक्तिगत कर्जाची निवड करताना खालील पैकी कोणत्या वैशिष्ट्यांचा तुम्हाला सर्वाधिक महत्त्व आहे?  
(कृपया फक्त ५ वैशिष्ट्ये निवडा)

☒ Checkboxes

☐ Processing Fee(प्रक्रिया शुल्क)

☐ Disbursal Time(वितरण वेळ)

#### Step 4: Phase 1 Analysis

The binary data was analyzed using the AI-Predictive Analytics Tool. The analysis helped identify the most influential factors affecting personal loan selection.

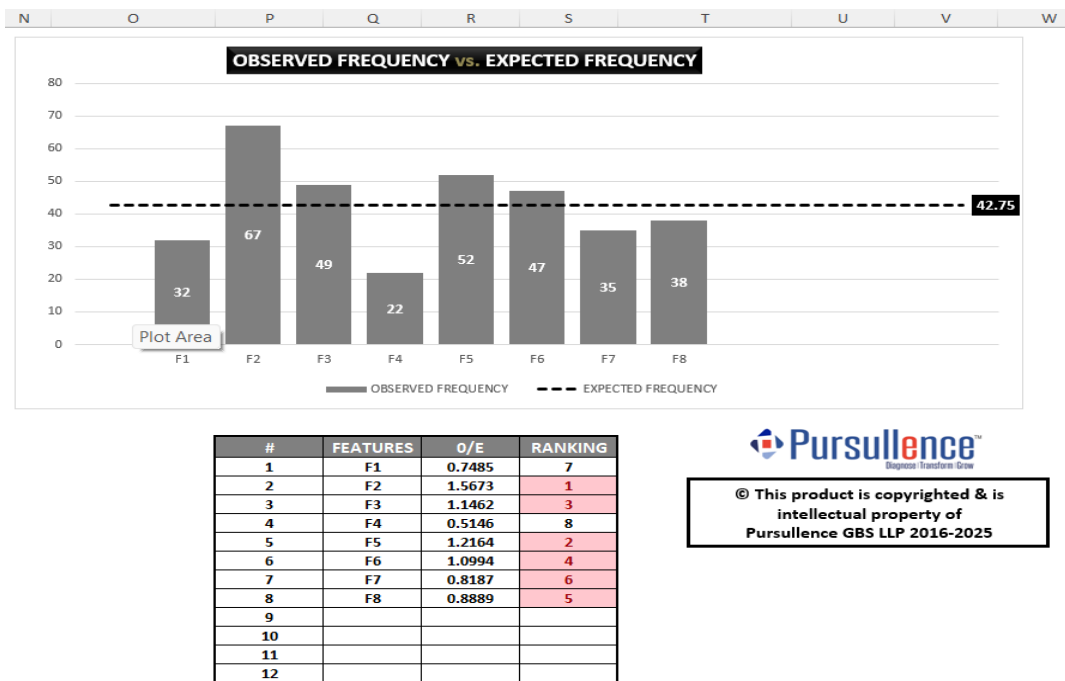
| OBSERVED FREQUENCY (OF) |     |     |       |
|-------------------------|-----|-----|-------|
| FEATURE / OFFER         | 1   | 0   | TOTAL |
| F1                      | 32  | 70  | 102   |
| F2                      | 67  | 35  | 102   |
| F3                      | 49  | 53  | 102   |
| F4                      | 22  | 80  | 102   |
| F5                      | 52  | 50  | 102   |
| F6                      | 47  | 55  | 102   |
| F7                      | 35  | 67  | 102   |
| F8                      | 38  | 64  | 102   |
|                         |     |     |       |
|                         |     |     |       |
|                         |     |     |       |
|                         |     |     |       |
| TOTAL                   | 342 | 474 | 816   |

|                 |        |
|-----------------|--------|
| TEST STATISTICS | 54.746 |
|-----------------|--------|

| EXPECTED FREQUENCY (EF) |       |       |       |
|-------------------------|-------|-------|-------|
| FEATURE / OFFER         | 1     | 0     | TOTAL |
| F1                      | 42.75 | 59.25 | 102   |
| F2                      | 42.75 | 59.25 | 102   |
| F3                      | 42.75 | 59.25 | 102   |
| F4                      | 42.75 | 59.25 | 102   |
| F5                      | 42.75 | 59.25 | 102   |
| F6                      | 42.75 | 59.25 | 102   |
| F7                      | 42.75 | 59.25 | 102   |
| F8                      | 42.75 | 59.25 | 102   |
|                         |       |       |       |
|                         |       |       |       |
|                         |       |       |       |
|                         |       |       |       |
| TOTAL                   | 342   | 474   | 816   |

|                      |         |
|----------------------|---------|
| P VALUE              | 0.00000 |
| GURRANTY / ASSURANCE | 100.00% |

The analysis indicated that **F2, F5, F3, F6, F8, and F7** were the six most preferred features and were therefore selected for further evaluation in Phase 2.



- 1] The statistical analysis produced a Test Statistic value of 54.746 and a P-Value of 0.00000, indicating significant variation in customer preferences.
- 2] The AI-Predictive Analytics Tool generated an Assurance Level of 100%, indicating high confidence in the obtained results.
- 3] Among the eight factors evaluated, F2 emerged as the most influential factor, followed by F5, F3, F6, F8, and F7.
- 4] These six factors were selected for detailed optimization in Phase 2.
- 5] Customer preferences can be effectively analyzed and ranked using predictive analytics techniques.

### Step 5: Phase 2 Questionnaire Design

Based on the significant factors identified in Phase 1, a second questionnaire was developed. Multiple alternatives were provided for each selected feature.

### Step 6: Phase 2 Data Collection

Responses were collected from approximately 200 participants through Google Forms.

## PERSONAL LOAN FEATURES & STRATEGIES PREFERENCE SURVEY

Thank you for participating in this survey. The purpose of this study is to understand customer preferences regarding Personal Loan product designs and service strategies. Your responses will be used for academic project analysis only.

\*\*\*

Q1. Which processing fee option would you prefer? \*

- ☐ 0% processing fee, slightly higher interest rate
- ☐ 1% processing fee, standard interest rate
- ☐ Flat ₹2,500 processing fee regardless of loan amount
- ☐ Processing fee refunded after 12 on-time EMIs

Q2. Which loan disbursal time would you prefer? \*

- ☐ Instant disbursal within 2 hours after approval
- ☐ Same-day disbursal
- ☐ Disbursal within 24–48 hours
- ☐ Disbursal within 3–5 working days with additional benefits

\*\*\*

Q3. Which documentation process would be most convenient for you? \*

- ☐ Fully digital KYC and document upload
- ☐ Digital application with physical verification
- ☐ Minimal documents but mandatory video verification
- ☐ Instant approval using bank statement access consent

Q4. What type of loan protection insurance policy would you prefer? \*

- ☐ Insurance bundled and mandatory
- ☐ Optional insurance at discounted premium
- ☐ Free first-year insurance included
- ☐ No insurance offered, lower EMI

...

Q5. Which flexible repayment tenure option would you prefer for a personal loan? \*

- ☐ Tenure from 1 year to 7 years
- ☐ Fixed tenure of 3 years with lower interest
- ☐ Flexible tenure change allowed once during loan
- ☐ Customized repayment schedule based on income pattern

...

Q6. Which interest structure would you prefer for your loan? \*

- ☐ Reducing balance interest rate
- ☐ Flat interest rate with predictable EMI
- ☐ Step-down interest rate over time
- ☐ Floating interest linked to market rate

## Step 7: Data Processing and Analysis

The Phase 2 responses were converted into binary format and analyzed using the AI-Predictive Analytics Tool to identify the most preferred feature combinations and optimized loan offerings.

|                                                                                 |    |             |                                    |        |                      |
|---------------------------------------------------------------------------------|----|-------------|------------------------------------|--------|----------------------|
| <b>Pursuence</b><br>© All Rights Reserved. Proprietary and highly Confidential. | #  | CATEGORY ID | CATEGORIES / ATTRIBUTES / FEATURES | OFFERS | ESTIMATED OFFER COST |
|                                                                                 | 1  | 1           | A                                  | A1     | Rs. 28,409.00        |
|                                                                                 | 2  | 1           | A                                  | A2     | Rs. 13,274.00        |
| PRODUCT STRATEGIES SPECTRUM                                                     | 3  | 1           | A                                  | A3     | Rs. 25,020.00        |
|                                                                                 | 4  | 1           | A                                  | A4     | Rs. 26,633.00        |
| PRODCT STRATEGY INGESTION LAYER                                                 | 5  | 2           | B                                  | B1     | Rs. 27,361.00        |
|                                                                                 | 6  | 2           | B                                  | B2     | Rs. 29,332.00        |
|                                                                                 | 7  | 2           | B                                  | B3     | Rs. 15,336.00        |
| PRODUCT STRATEGY CRUCIBLE                                                       | 8  | 2           | B                                  | B4     | Rs. 29,514.00        |
|                                                                                 | 9  | 3           | C                                  | C1     | Rs. 13,611.00        |
|                                                                                 | 10 | 3           | C                                  | C2     | Rs. 12,834.00        |
| ARTIFICIAL CHRONOS ENGINE                                                       | 11 | 3           | C                                  | C3     | Rs. 31,073.00        |
|                                                                                 | 12 | 3           | C                                  | C4     | Rs. 12,096.00        |
|                                                                                 | 13 | 4           | D                                  | D1     | Rs. 31,187.00        |
| ARTIFICIAL INFERENCE AUGUR                                                      | 14 | 4           | D                                  | D2     | Rs. 22,613.00        |
|                                                                                 | 15 | 4           | D                                  | D3     | Rs. 30,605.00        |
|                                                                                 | 16 | 4           | D                                  | D4     | Rs. 34,990.00        |
| PRODUCT DESIGN ARCHITECT                                                        | 17 | 5           | E                                  | E1     | Rs. 28,947.00        |
|                                                                                 | 18 | 5           | E                                  | E2     | Rs. 16,876.00        |
|                                                                                 | 19 | 5           | E                                  | E3     | Rs. 33,915.00        |
| PRODUCT DESIGN OPTIMISER                                                        | 20 | 5           | E                                  | E4     | Rs. 16,046.00        |
|                                                                                 | 21 | 6           | F                                  | F1     | Rs. 17,065.00        |
|                                                                                 | 22 | 6           | F                                  | F2     | Rs. 17,012.00        |
|                                                                                 | 23 | 6           | F                                  | F3     | Rs. 16,562.00        |
|                                                                                 | 24 | 6           | F                                  | F4     | Rs. 31,027.00        |

Respective Est. Offer Cost for each of 24 offers.

|                                                                                 |                |         |        |     |                |                  |                 |                                 |                      |               |                    |
|---------------------------------------------------------------------------------|----------------|---------|--------|-----|----------------|------------------|-----------------|---------------------------------|----------------------|---------------|--------------------|
| <b>Pursuence</b><br>© All Rights Reserved. Proprietary and highly Confidential. | Strategy/Offer | YES (s) | NO (f) | n   | sample p (s/n) | $\alpha = (s+1)$ | $\beta = (f+1)$ | $U = \alpha / (\alpha + \beta)$ | Prob $\theta > \tau$ | Cost          | $\tau$ (min pref.) |
|                                                                                 | A - A1         | 29      | 171    | 200 | 0.15           | 30               | 172             | 0.1485                          | 0.0002               | Rs. 28,409.00 | 0.25               |
| PRODUCT STRATEGIES SPECTRUM                                                     | A - A2         | 82      | 118    | 200 | 0.41           | 83               | 119             | 0.4109                          | 1.0000               | Rs. 13,274.00 | 0.25               |
|                                                                                 | A - A3         | 67      | 133    | 200 | 0.34           | 68               | 134             | 0.3366                          | 0.9968               | Rs. 25,020.00 | 0.25               |
| PRODCT STRATEGY INGESTION LAYER                                                 | A - A4         | 22      | 178    | 200 | 0.11           | 23               | 179             | 0.1139                          | 0.0000               | Rs. 26,633.00 | 0.25               |
|                                                                                 | B - B1         | 65      | 135    | 200 | 0.33           | 66               | 136             | 0.3267                          | 0.9923               | Rs. 27,361.00 | 0.25               |
| PRODUCT STRATEGY CRUCIBLE                                                       | B - B2         | 63      | 137    | 200 | 0.32           | 64               | 138             | 0.3168                          | 0.9828               | Rs. 29,332.00 | 0.25               |
|                                                                                 | B - B3         | 51      | 149    | 200 | 0.26           | 52               | 150             | 0.2574                          | 0.5857               | Rs. 15,336.00 | 0.25               |
| ARTIFICIAL CHRONOS ENGINE                                                       | B - B4         | 21      | 179    | 200 | 0.11           | 22               | 180             | 0.1089                          | 0.0000               | Rs. 29,514.00 | 0.25               |
|                                                                                 | C - C1         | 49      | 151    | 200 | 0.25           | 50               | 152             | 0.2475                          | 0.4567               | Rs. 13,611.00 | 0.25               |
| ARTIFICIAL INFERENCE AUGUR                                                      | C - C2         | 65      | 135    | 200 | 0.33           | 66               | 136             | 0.3267                          | 0.9923               | Rs. 12,834.00 | 0.25               |
|                                                                                 | C - C3         | 59      | 141    | 200 | 0.30           | 60               | 142             | 0.2970                          | 0.9321               | Rs. 31,073.00 | 0.25               |
| PRODUCT DESIGN ARCHITECT                                                        | C - C4         | 27      | 173    | 200 | 0.14           | 28               | 174             | 0.1386                          | 0.0000               | Rs. 12,096.00 | 0.25               |
|                                                                                 | D - D1         | 38      | 162    | 200 | 0.19           | 39               | 163             | 0.1931                          | 0.0252               | Rs. 31,187.00 | 0.25               |
| PRODUCT DESIGN OPTIMISER                                                        | D - D2         | 88      | 112    | 200 | 0.44           | 89               | 113             | 0.4406                          | 1.0000               | Rs. 22,613.00 | 0.25               |
|                                                                                 | D - D3         | 42      | 158    | 200 | 0.21           | 43               | 159             | 0.2129                          | 0.1018               | Rs. 30,605.00 | 0.25               |
|                                                                                 | D - D4         | 32      | 168    | 200 | 0.16           | 33               | 169             | 0.1634                          | 0.0013               | Rs. 34,990.00 | 0.25               |
|                                                                                 | E - E1         | 42      | 158    | 200 | 0.21           | 43               | 159             | 0.2129                          | 0.1018               | Rs. 28,947.00 | 0.25               |
|                                                                                 | E - E2         | 84      | 116    | 200 | 0.42           | 85               | 117             | 0.4208                          | 1.0000               | Rs. 16,876.00 | 0.25               |
|                                                                                 | E - E3         | 54      | 146    | 200 | 0.27           | 55               | 147             | 0.2723                          | 0.7578               | Rs. 33,915.00 | 0.25               |
|                                                                                 | E - E4         | 20      | 180    | 200 | 0.10           | 21               | 181             | 0.1040                          | 0.0000               | Rs. 16,046.00 | 0.25               |
|                                                                                 | F - F1         | 56      | 144    | 200 | 0.28           | 57               | 145             | 0.2822                          | 0.8456               | Rs. 17,065.00 | 0.25               |
|                                                                                 | F - F2         | 84      | 116    | 200 | 0.42           | 85               | 117             | 0.4208                          | 1.0000               | Rs. 17,012.00 | 0.25               |
|                                                                                 | F - F3         | 42      | 158    | 200 | 0.21           | 43               | 159             | 0.2129                          | 0.1018               | Rs. 16,562.00 | 0.25               |
|                                                                                 | F - F4         | 18      | 182    | 200 | 0.09           | 19               | 183             | 0.0941                          | 0.0000               | Rs. 31,027.00 | 0.25               |

The Probability of  $\theta > \tau$  value represents the probability of actual success rate/attractiveness across entire population more than  $\tau$  i.e. expected success rate/attractiveness for a given offer.

| Strategy/Offer | YES (s) | NO (f) | n    | Sample p<br>(s/n) | $\alpha=(s+1)$ | $\beta=(f+1)$ | Posterior<br>$\mu=\alpha/(\alpha+\beta)$ | Prob $\theta>\tau$ | Cost          | Value/Cost | $\tau$ (min<br>pref.) |
|----------------|---------|--------|------|-------------------|----------------|---------------|------------------------------------------|--------------------|---------------|------------|-----------------------|
| A - A1         | 759     | 4241   | 5000 | 0.1518            | 760            | 4242          | 0.15194                                  | 0.000              | Rs. 28,409.00 | 0.00000535 | 0.25                  |
| A - A2         | 2120    | 2880   | 5000 | 0.424             | 2121           | 2881          | 0.42403                                  | 1.000              | Rs. 13,274.00 | 0.00003194 | 0.25                  |
| A - A3         | 1720    | 3280   | 5000 | 0.344             | 1721           | 3281          | 0.34406                                  | 1.000              | Rs. 25,020.00 | 0.00001375 | 0.25                  |
| A - A4         | 527     | 4473   | 5000 | 0.1054            | 528            | 4474          | 0.10556                                  | 0.000              | Rs. 26,633.00 | 0.00000396 | 0.25                  |
| B - B1         | 1600    | 3400   | 5000 | 0.32              | 1601           | 3401          | 0.32007                                  | 1.000              | Rs. 27,361.00 | 0.00001170 | 0.25                  |
| B - B2         | 1534    | 3466   | 5000 | 0.3068            | 1535           | 3467          | 0.30688                                  | 1.000              | Rs. 29,332.00 | 0.00001046 | 0.25                  |
| B - B3         | 1269    | 3731   | 5000 | 0.2538            | 1270           | 3732          | 0.25390                                  | 0.736              | Rs. 15,336.00 | 0.00001656 | 0.25                  |
| B - B4         | 552     | 4448   | 5000 | 0.1104            | 553            | 4449          | 0.11056                                  | 0.000              | Rs. 29,514.00 | 0.00000375 | 0.25                  |
| C - C1         | 1256    | 3744   | 5000 | 0.2512            | 1257           | 3745          | 0.25130                                  | 0.582              | Rs. 13,611.00 | 0.00001846 | 0.25                  |
| C - C2         | 1634    | 3366   | 5000 | 0.3268            | 1635           | 3367          | 0.32687                                  | 1.000              | Rs. 12,834.00 | 0.00002547 | 0.25                  |
| C - C3         | 1484    | 3516   | 5000 | 0.2968            | 1485           | 3517          | 0.29688                                  | 1.000              | Rs. 31,073.00 | 0.00000955 | 0.25                  |
| C - C4         | 712     | 4288   | 5000 | 0.1424            | 713            | 4289          | 0.14254                                  | 0.000              | Rs. 12,096.00 | 0.00001178 | 0.25                  |
| D - D1         | 937     | 4063   | 5000 | 0.1874            | 938            | 4064          | 0.18752                                  | 0.000              | Rs. 31,187.00 | 0.00000601 | 0.25                  |
| D - D2         | 2199    | 2801   | 5000 | 0.4398            | 2200           | 2802          | 0.43982                                  | 1.000              | Rs. 22,613.00 | 0.00001945 | 0.25                  |
| D - D3         | 1081    | 3919   | 5000 | 0.2162            | 1082           | 3920          | 0.21631                                  | 0.000              | Rs. 30,605.00 | 0.00000707 | 0.25                  |
| D - D4         | 803     | 4197   | 5000 | 0.1606            | 804            | 4198          | 0.16074                                  | 0.000              | Rs. 34,990.00 | 0.00000459 | 0.25                  |
| E - E1         | 1077    | 3923   | 5000 | 0.2154            | 1078           | 3924          | 0.21551                                  | 0.000              | Rs. 28,947.00 | 0.00000745 | 0.25                  |
| E - E2         | 2077    | 2923   | 5000 | 0.4154            | 2078           | 2924          | 0.41543                                  | 1.000              | Rs. 16,876.00 | 0.00002462 | 0.25                  |
| E - E3         | 1380    | 3620   | 5000 | 0.276             | 1381           | 3621          | 0.27609                                  | 1.000              | Rs. 33,915.00 | 0.00000814 | 0.25                  |
| E - E4         | 515     | 4485   | 5000 | 0.103             | 516            | 4486          | 0.10316                                  | 0.000              | Rs. 16,046.00 | 0.00000643 | 0.25                  |
| F - F1         | 1485    | 3515   | 5000 | 0.297             | 1486           | 3516          | 0.29708                                  | 1.000              | Rs. 17,065.00 | 0.00001741 | 0.25                  |
| F - F2         | 2110    | 2890   | 5000 | 0.422             | 2111           | 2891          | 0.42203                                  | 1.000              | Rs. 17,012.00 | 0.00002481 | 0.25                  |
| F - F3         | 1100    | 3900   | 5000 | 0.22              | 1101           | 3901          | 0.22011                                  | 0.000              | Rs. 16,562.00 | 0.00001329 | 0.25                  |
| F - F4         | 465     | 4535   | 5000 | 0.093             | 466            | 4536          | 0.09316                                  | 0.000              | Rs. 31,027.00 | 0.00000300 | 0.25                  |

| FEATURE-1 | FEATURE-2 | FEATURE-3 | FEATURE-4 | FEATURE-5 | FEATURE-6 | UA           | UB           | UC           | UD           | UE           | UF           | TOTAL       | Rank |
|-----------|-----------|-----------|-----------|-----------|-----------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|------|
| A1        | B3        | C2        | D2        | E2        | F2        | -8.40363E-06 | 5.94024E-06  | 9.15136E-06  | 1.01689E-05  | 1.2959E-05   | 1.01805E-05  | 3.99963E-05 | 83   |
| A2        | B1        | C1        | D1        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 2.14532E-06  | -3.26824E-06 | 1.2959E-05   | 1.01805E-05  | 4.12917E-05 | 71   |
| A2        | B1        | C1        | D2        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | 2.14532E-06  | 1.01689E-05  | 1.2959E-05   | 2.78143E-06  | 4.73298E-05 | 29   |
| A2        | B1        | C1        | D2        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 2.14532E-06  | 1.01689E-05  | 1.2959E-05   | 1.01805E-05  | 5.47288E-05 | 7    |
| A2        | B1        | C1        | D2        | E2        | F3        | 1.81925E-05  | 1.08263E-06  | 2.14532E-06  | 1.01689E-05  | 1.2959E-05   | -1.33719E-06 | 4.32111E-05 | 54   |
| A2        | B1        | C1        | D3        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 2.14532E-06  | -2.21325E-06 | 1.2959E-05   | 1.01805E-05  | 4.23467E-05 | 59   |
| A2        | B1        | C1        | D4        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 2.14532E-06  | -4.6874E-06  | 1.2959E-05   | 1.01805E-05  | 3.98725E-05 | 85   |
| A2        | B1        | C2        | D1        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -3.26824E-06 | 1.2959E-05   | 2.78143E-06  | 4.08987E-05 | 75   |
| A2        | B1        | C2        | D1        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -3.26824E-06 | 1.2959E-05   | 1.01805E-05  | 4.82977E-05 | 25   |
| A2        | B1        | C2        | D2        | E1        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -4.21277E-06 | 1.01689E-05  | 1.01805E-05  | 4.45631E-05 | 47   |
| A2        | B1        | C2        | D2        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | 1.01689E-05  | 1.2959E-05   | 2.78143E-06  | 5.43358E-05 | 8    |
| A2        | B1        | C2        | D2        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | 1.01689E-05  | 1.2959E-05   | 1.01805E-05  | 6.17349E-05 | 2    |
| A2        | B1        | C2        | D2        | E2        | F3        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | 1.01689E-05  | 1.2959E-05   | -1.33719E-06 | 5.02172E-05 | 18   |
| A2        | B1        | C2        | D2        | E2        | F4        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | 1.01689E-05  | 1.2959E-05   | -1.16247E-05 | 3.99296E-05 | 84   |
| A2        | B1        | C2        | D2        | E3        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | 1.01689E-05  | -3.51725E-06 | 1.01805E-05  | 4.52586E-05 | 43   |
| A2        | B1        | C2        | D2        | E4        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | 1.01689E-05  | -5.22894E-06 | 1.01805E-05  | 4.3547E-05  | 50   |
| A2        | B1        | C2        | D3        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -2.21325E-06 | 1.2959E-05   | 2.78143E-06  | 4.19537E-05 | 65   |
| A2        | B1        | C2        | D3        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -2.21325E-06 | 1.2959E-05   | 1.01805E-05  | 4.93527E-05 | 21   |
| A2        | B1        | C2        | D4        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -4.6874E-06  | 1.2959E-05   | 2.78143E-06  | 3.94795E-05 | 88   |
| A2        | B1        | C2        | D4        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | 9.15136E-06  | -4.6874E-06  | 1.2959E-05   | 1.01805E-05  | 4.68786E-05 | 32   |
| A2        | B1        | C3        | D2        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | -6.76333E-06 | 1.01689E-05  | 1.2959E-05   | 2.78143E-06  | 3.84211E-05 | 99   |
| A2        | B1        | C3        | D2        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | -6.76333E-06 | 1.01689E-05  | 1.2959E-05   | 1.01805E-05  | 4.58202E-05 | 39   |
| A2        | B1        | C4        | D2        | E2        | F1        | 1.81925E-05  | 1.08263E-06  | -4.53334E-06 | 1.01689E-05  | 1.2959E-05   | 2.78143E-06  | 4.06511E-05 | 77   |
| A2        | B1        | C4        | D2        | E2        | F2        | 1.81925E-05  | 1.08263E-06  | -4.53334E-06 | 1.01689E-05  | 1.2959E-05   | 1.01805E-05  | 4.80502E-05 | 28   |
| A2        | B2        | C1        | D1        | E2        | F2        | 1.81925E-05  | -1.53275E-07 | 2.14532E-06  | -3.26824E-06 | 1.2959E-05   | 1.01805E-05  | 4.00558E-05 | 82   |
| A2        | B2        | C1        | D2        | E2        | F1        | 1.81925E-05  | -1.53275E-07 | 2.14532E-06  | 1.01689E-05  | 1.2959E-05   | 2.78143E-06  | 4.69430E-05 | 38   |

Based on the algorithm for utility scores, under a 2nd layer of a statistical modeling , system has analyzed the data and predicted the "RANKING" for each of 4096 combinations. There are 4096 combinations for 4 offers and 6 features , 4\*4\*4\*4\*4\*4=4096.

In tool , Ranking goes till total of 4096 rows.

System has given top 100 Ranked combinations of offers across all 6 features.

|                         |    |    |    |    |    |    |
|-------------------------|----|----|----|----|----|----|
| 80                      | A2 | B3 | C3 | D3 | E2 | F2 |
| 81                      | A2 | B3 | C4 | D1 | E2 | F2 |
| 82                      | A2 | B3 | C4 | D2 | E2 | F1 |
| 83                      | A2 | B3 | C4 | D2 | E2 | F2 |
| 84                      | A2 | B3 | C4 | D2 | E2 | F3 |
| 85                      | A2 | B3 | C4 | D3 | E2 | F2 |
| 86                      | A2 | B4 | C1 | D2 | E2 | F1 |
| 87                      | A2 | B4 | C1 | D2 | E2 | F2 |
| 88                      | A2 | B4 | C2 | D1 | E2 | F2 |
| 89                      | A2 | B4 | C2 | D2 | E2 | F1 |
| 90                      | A2 | B4 | C2 | D2 | E2 | F2 |
| 91                      | A2 | B4 | C2 | D2 | E2 | F3 |
| 92                      | A2 | B4 | C2 | D3 | E2 | F2 |
| 93                      | A2 | B4 | C2 | D4 | E2 | F2 |
| 94                      | A2 | B4 | C4 | D2 | E2 | F2 |
| 95                      | A3 | B1 | C2 | D2 | E2 | F2 |
| 96                      | A3 | B2 | C2 | D2 | E2 | F2 |
| 97                      | A3 | B3 | C1 | D2 | E2 | F2 |
| 98                      | A3 | B3 | C2 | D2 | E2 | F1 |
| 99                      | A3 | B3 | C2 | D2 | E2 | F2 |
| 100                     | A4 | B3 | C2 | D2 | E2 | F2 |
| SIGNIFICANT COMBINATION |    |    |    |    |    |    |
|                         | A2 | B3 | C2 | D2 | E2 | F2 |

Algorithm in Pursullence Ai Predictive Analytics tool has selected the

'OPTIMUM' combination of best offers/strategies of each of 6 features. This is statistically designed Optimum Product liked by customers.

**A2 B3 C2 D2 E2 F2** are found to be the most significant combination among all the studied ones.

## Step 8: Interpretation and Recommendations

The analytical outputs were interpreted and used to develop recommendations for designing customer-centric personal loan products.



## **Implementation Strategies**

Based on the findings of the study, financial institutions can implement customer-centric personal loan products by focusing on the features most preferred by customers. The identified high-priority loan attributes should be incorporated into product design and marketing strategies.

Banks and financial institutions can utilize customer preference data to:

- Design loan products aligned with customer expectations.
- Improve loan approval and disbursal processes.
- Offer flexible repayment options.
- Develop competitive interest rate structures.
- Enhance customer satisfaction and retention.
- Use predictive analytics for continuous product optimization.

The implementation of these strategies can help institutions improve product acceptance and gain a competitive advantage in the personal loan market.

## Limitations and Drawbacks

- The study was based on a limited sample of respondents.
- Responses were collected through online surveys and may contain respondent bias.
- Customer preferences may change over time due to economic and market conditions.
- The study focused only on selected personal loan attributes.
- Results are dependent on the accuracy and honesty of participant responses.
- Findings may not fully represent the preferences of the entire population.

## Observations

- Customers consider multiple factors before selecting a personal loan product.
- Interest-related factors and repayment flexibility significantly influence decision-making.
- Convenience-oriented features such as faster disbursal and simplified documentation are highly valued.
- Certain loan features received considerably higher preference scores than others.
- Customer preferences can be effectively analyzed using predictive analytics techniques.
- Data-driven analysis provides meaningful insights for product optimization and business decision-making.
- **A2 B3 C2 D2 E2 F2** are found to be the most significant combination among all the studied ones.

## **Conclusion**

The study successfully identified the major factors influencing personal loan selection and analyzed customer preferences regarding various loan features. Through survey-based data collection and predictive analytics, valuable insights were obtained regarding customer expectations from personal loan products.

The results indicate that customers prefer loan products that offer affordability, convenience, flexibility, and transparency. The application of the AI-Predictive Analytics Tool helped in identifying preferred feature combinations and optimizing product offerings. The study demonstrates the usefulness of data-driven approaches in designing customer-centric financial products and improving business decision-making.

## **Future Enhancements**

- Increase the sample size to obtain broader customer representation.
- Include additional loan-related factors such as customer service quality and digital banking features.
- Conduct demographic-based analysis to study preference variations across age groups, occupations, and income levels.
- Apply advanced machine learning techniques for deeper predictive analysis.
- Perform periodic studies to capture changing customer preferences and market trends.
- Extend the framework to other financial products such as home loans, vehicle loans, and credit cards.

## **User Manual**

### **Step 1**

Create and distribute the questionnaire using Google Forms.

### **Step 2**

Collect responses from participants and export the data to Microsoft Excel.

### **Step 3**

Convert the collected responses into the required binary (0–1) format.

### **Step 4**

Input the processed data into the AI-Predictive Analytics Tool.

### **Step 5**

Execute the analysis to generate frequency distributions, rankings, and optimization results.

### **Step 6**

Interpret the analytical outputs and prepare recommendations based on the findings.

### **Step 7**

Use the generated insights for product design, strategy formulation, and decision-making.

## **References**

- ☐ Google Forms Documentation.
- ☐ Microsoft Excel Documentation.
- ☐ Pursullence AI-Predictive Analytics Tool User Manual
- ☐ Books on Marketing Research and Business Analytics.
- ☐ Academic articles on Customer Preference Analysis and Predictive Analytics.

